SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

ENG301

Timed Online Assignment – July Semester 2021

Microprocessor Programming

Tuesday, 16 November 2021 **4:00 pm – 6:30 pm**

Time allowed: 2.5 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **FOUR (4)** questions and comprises **SEVEN (7)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOACourse Code, Your name and Student PI No. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT space. Example, TOAENG301_RaphaelLee_T1923161 (omit D/O, S/O).

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6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a Microsoft Word file saved in .doc or .docx format. All answers are to be typed in Times New Roman font, size 12 and single spacing. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. Flowcharts and graphs may be scanned or photographed and embedded in the Word file in the right sequence provided it does not exceed the file size limit of 500 MB. All uploaded handwritten answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

7. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

8. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

Answer all questions. (Total 100 marks)

Question 1

(a) Examine the principles and basic concepts of pipeline design and answer the following.

(i) Execute the assembly instructions in the same order as they are found in Table Q1(a)(i) and fill in the table by indicating the stage of the standard 5-stage pipeline that the instruction will be in during the clock cycles. If an instruction is not in any stage during a cycle, simply leave that box blank. Indicate the stages of the pipeline using: F, D, X, M, and W. If there is gap in stages due to a data hazard, make sure to indicate the data hazard using d*. *Note; The pipeline has no bypassing.*

	Clock Cycle														
Instructions	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
SUB R2, R3, R1															
LDR R5, [R2]															
STR [R4], R5															
ADD R0, R4, R1															

Table Q1(a)(i)

(8 marks)

(ii) Fill in the Table Q1(a)(ii) when the pipeline design has full bypassing.

	Clock Cycle														
Instructions	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
SUB R2, R3, R1															
LDR R5, [R2]															
STR [R4], R5															
ADD R0, R4, R1															

Table Q1(a)(ii)

(8 marks)

(iii) Appraise the percentage improvement in the performance of the code fragment given in Tables Q1(a)(i) and Q1(a)(ii) due to bypassing.

(3 marks)

(b) An instruction pipeline consists of 4 stages: Fetch (F), Decode field (D), Execute (E), and Result-Write (W). The 5 instructions in a certain instruction sequence need these stages for the different number of clock cycles as shown in Table Q1(b). Assuming no branch and data hazard in the 5 instructions and each stage can only execute 1 instruction at a time, find the number of clock cycles needed to perform the 5 instructions.

	No of clock cycles needed			
Instruction	F	D	E	W
1	1	2	1	1
2	1	2	2	1
3	2	1	3	2
4	1	3	2	1
5	1	2	1	2

Table Q1(b)

(6 marks)

Question 2

(a) The instruction code given below is run on an ARM processor.

```
main:
        SUB R3, R3, #1
init:
        MOV R0, #0
        MOV R1, #1
        MOV R2, #0
loop:
        LDR R5, [R4, R0, LSL #2]
        LDR R6, [R4, R1, LSL #2]
        CMP R5, R6
        BGT loop2
rtn:
        ADD R0, R0, #1
        ADD R1, R1, #1
        CMP R0, R3
        BLT loop
        CMP R2, #0
        BGT init
        B exit
loop2:
        ADD R2, R2, #1
        STR R6, [R4, R0, LSL #2]
        STR R5, [R4, R1, LSL #2]
        B rtn
exit:
        MOV R0, #0
        MOV PC, LR
```

(i) Assume that register R4 holds the base address in memory for an array of integers of length N, register R3 contains the length of the array (i.e. N), and registers R0, R1, and R2 are counters. Analyse the code and explain the function of each line of code by inserting appropriate comments.

(14 marks)

(ii) Given that R3 = 5 and R4 = 0x8070. The values stored in the memory Block before the execution of the code is provided in Table Q2(a)(ii). Obtain the values stored in the memory block after the execution of the code.

MEM Location	Data
0x8070	36
0x8074	36
0x8078	29
0x807C	42
0x8080	13

Table Q2(a)(ii)

(3 marks)

(b) Consider the following comparison instruction:

TST R0, R1, ASR #1 ; R0 = 0x12345678 and R1 = 0xDB97530F

(i) Appraise the value of the condition flags (N, Z, C, V) after the execution of the instruction. Use the values of R0 and R1 provided in the instruction comment.

(4 marks)

(ii) The conditional branches are BHS, BLO, BLT and BPL. From the condition flag bits appraised in Question 2(b)(i), determine which of the above conditional branches will be executed.

(4 marks)

Question 3

(a) Demonstrate your knowledge in ARM programming languages by translating the following code into an equivalent and optimised sequence of ARM instructions.

```
int my_func(int x, int y) {
       if (x >= 0 && y >= 0)
              return (x + y);
       else if (x >= 0 && y < 0)
              return (x - y);
       else if (x < 0 && y >= 0)
              return (y - x);
       else
              return (x - y);
 }
```

(12 marks)

(b) Analyse the contents of the stack as well as any change in the position of the Stack Pointer after each of the following instruction is executed in Table Q3(b). Assume R0 contains the value of 0xFFFFFFFF and Stack Pointer is pointing to 0x008010.
(*NOTE: All four questions are independent from each other*.)

Memory Address	Memory Content
0x00008000	
0x00008004	
0x00008008	
0x0000800C	
0x00008010	001A2203
0x00008014	F0034A21
0x00008018	0F32D360

Table Q3(b)

(i) STR R0, [SP, # -8]
(ii) STR R0, [SP, # -8] !
(iii) ADD SP, SP, #4
(iv) LDR R0, [SP], #4

(8 marks)

(c) Explain the function of co-processor and give an example of a co-processor.
(5 marks)

Question 4

(a) Demonstrate your understanding on ARM system design by answering following question

(i) In an ARM System, an input device is memory mapped into a 8-bit memory map location. InPort is defined as pointing to the address location of the device. A push button is connected to the rightmost bit of the device and is wired such that the input bit sees a value of 0 when push button is not pressed and sees a value of 1 when pressed. The other 7 bits are connected to other input device which we are not interested in but will still be read in. Design a subroutine called WAIT_Button that polls the input device until the Push button is pressed.
(10 marks)

(ii) Examine the advantages and disadvantages in the above design. Propose how one can improve the above system design.
(5 marks)

(b) Demonstrate your understanding of memory system by designing a basic ARM memory system with the following information

An ARM memory system has 4 physical 4Kx8 EPROM memory located at 0000h. There are also 4 physical 4Kx8 RAM memory modules. The RAM module starts from C000h.

(i) Sketch the memory map of the system.

(5 marks)

(ii) Sketch the memory system showing the

- Memory Modules
- ARM block
- Control Block
- Relevant address and control lines

(5 marks)

----- END OF PAPER -----